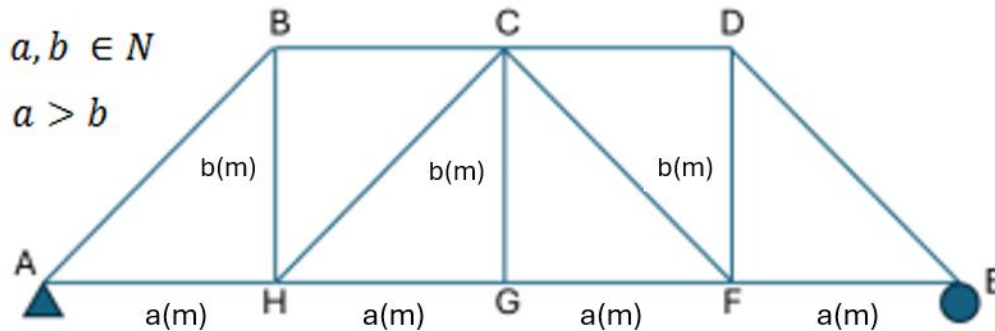


Project 3

The goal of this project is to create a Truss and run a vibration analysis (FEA modal study) in SOLIDWORKS to determine the natural frequencies and mode shapes of the structure.



1. Set Units to MKS

- Click the Unit System dropdown at the bottom-right corner of the SolidWorks window.
- Select MKS (meter, kilogram, second).

OR

- Go to the top menu and click Options (gear icon).
- In the System Options dialog, select Document Properties > Units.
- Set the unit system to MKS.

2. Create a Sketch on the Front Plane

- In the Feature Manager Design Tree on the left, right-click Front Plane.
- Click Sketch from the context menu that appears. This opens a new sketch on the front plane.

3. Draw the Truss Members

- Select the Line tool from the Sketch tab in the Command Manager toolbar at the top.
- Draw lines representing each truss member: AH, HG, GF, FE, ED, DC, CB, BA, BH, HC, CG, CF, FD.
- Use dimensions as per the project drawing to size members accurately.

4. Exit the Sketch

- Click the Exit Sketch button at the top right of the graphics area (near the confirmation corner).
- Alternatively, right-click in the graphics area and select Exit Sketch.

5. Enable the Weldments Tab if Not Visible

- Right-click anywhere on the tabs bar at the top (where tabs like Features, Sketch, etc. are located).
- Click Tabs.
- Ensure the Weldments checkbox is checked to enable the Weldments tab.

6. Add Structural Members

- Click the Weldments tab on the Command Manager.
- Click Structural Member in the Weldments toolbar.

7. Select Profile Standard and Size

- In the Structural Member Property Manager on the left, set:
 - o Standard to ISO.
 - o Type to any except Rectangular Tube.
 - o Size to any.

8. Select Outside Beams

- In the graphics area, click to select the lines representing the outside beams of the truss.
- This will form Group 1.

9. Create a New Group and Select the Middle Beam

- In the Property Manager, click New Group.
- Select the middle beam line.
- This forms Group 2.

10. Create a New Group and Select Two Main Diagonal Beams

- Click New Group again.
- Select the two main diagonal beams.
- This forms Group 3.

11. Create a New Group and Select Left Side Beams

- Click New Group.
- Select the remaining beams on the left side.
- This forms Group 4.

12. Create a New Group and Select Right Side Beams

- Click New Group.
- Select the remaining beams on the right side.
- This forms Group 5.
- Finally, click the green checkmark in the Property Manager to exit the Structural Member feature.

13. Rename Beams (Optional)

- In the Feature Manager Design Tree, expand the Weldments folder.
- Right-click any beam feature to rename it for easier identification.

14. Switch to the Simulation Tab

- Click the Simulation tab in the Command Manager toolbar.
- If not visible, enable it via Tools > Add-Ins > SOLIDWORKS Simulation.

15. Create a Frequency Study

- Turn on Simulation: Tools > Add-Ins... > check SolidWorks Simulation.
- Simulation tab > New Study > Frequency > name the study Truss_Frequency.

16. Convert to Truss Elements

- In the design tree, click the arrow next to the part to expand the cut list.
- Expand the cut list, select the desired structural member, then right-click and choose Edit Definition.
- Change the member type from Beam (default) to Truss.

17. Apply Supports

a) Pin (Left Bottom Joint)

- Right-click Fixtures > Fixed Geometry.
- Select the leftmost bottom joint.
- This acts as the pin support.

b) Roller (Right Bottom Joint)

- Right-click Fixtures > Fixed Geometry.
- Select the rightmost bottom joint.
- Restrain: $U_y = 0$, $U_z = 0$.
- Leave U_x free (roller behavior in X).

c) All Other Joints (Normal to Plane)

- Right-click Fixtures > Fixed Geometry > Use Reference Geometry.

- Select the Front Plane.
- Apply to all remaining truss joints.
- Restrain: $U_z = 0$ only.
- Keeps the model 2D in-plane.

18. Assign Material

- Right-click the part > Apply/Edit Material.
- Select the same material from Project 2 (e.g., Alloy Steel).
- Confirm the density is correct.

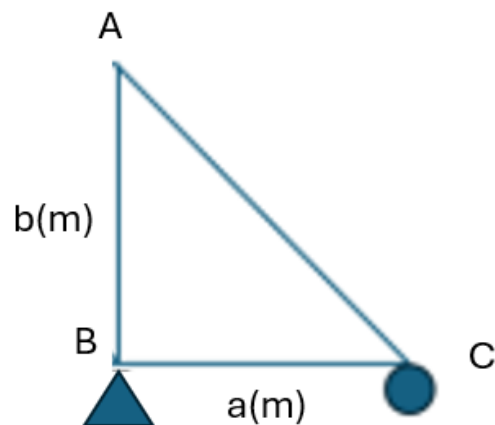
19. Study Properties & Mesh

- Right-click the study name > Properties... > set Number of frequencies = 10.
- Right-click Mesh > Create Mesh > accept defaults.
- If long members look too coarse, Split them for refinement.

20. Run the Study

- Click Run (green triangle).
- After solving, right-click Results > List Resonant Frequencies > copy the table.
- Right-click each Mode Shape (1-10) > Show/Animate.
- Display Deformed + Undeformed and adjust scale if needed.
- Save images and/or animations for the report.

PART B.



Find the first, second and third natural frequencies of the system above and compare with analytical solutions.

Report Content

Figures

The figures in the report should clearly show the geometry, supports, mesh, and vibration results.

- The model with dimensions should be presented in a figure so the overall geometry is clear.
- The model with boundary conditions should be shown to highlight the pin support, the roller support, and the restraints normal to the plane.
- A mesh figure should be included to display how the truss is discretized for analysis.
- Mode shape figures should be provided for each frequency, with the maximum and minimum displacements indicated.
- A resonant frequencies table should be included to summarize the results in numerical form.
- An animation video should be created to illustrate how the truss deforms in each mode shape.

Analysis and Interpretation

The analysis and interpretation should explain how the mode shapes and frequencies relate to the structural response of the truss.

- The deformation analysis should describe how the truss deforms in each mode, such as global sway, diagonal stretching, or symmetric oscillations.
- The natural frequency analysis should explain why natural frequency is important in engineering and should discuss how the obtained results affect the design of the truss.
- Brief description of its real-world relevance.

Rubric

Category	Criteria	Points
1. Plots & Figures		10 pts
	Model with Dimensions	2 pts
	Model with boundary condition shown	1 pt
	Mesh of the model	1 pts
	All mode shapes with maximum and minimum frequency (separate figures)	3 pts
	Table with resonant frequencies of all mode shapes	2 pts
	Animation videos of all modes	1 pts
2. Results Analysis		10 pts
	Brief description of its real-world relevance	1 pts
	Analysis of how the structure deforms in each mode.	2.0 pts
	Analysis of natural frequency and its importance.	2.0 pts
	Part B. Analysis and Comparison	5 pts
4. Report Quality		5 pts
	Proper formatting (Introduction, Methods, Results, Discussion sections)	3 pts
	Professional figures with captions and labels	2 pts
	Total	25 pts